# Chip usage data of the Tianyan-287 Inference Executions

The following tables contain the raw data of the quantum circuit executions of the self-attention layer and the feed-forward (FFN) layer of our quantum model on Tianyan-287. Because of the limitations of the depth of quantum circuits of Tianyan chips, the 2 layers are split into fragments to execute. Specifically, the self-attention layer is split into 24 segments, and the feed-forward layer is split into 10. Each task contains 10 shots of circuit executions. We repeat all the experiments 3 times. Chip usage time listed in the tables takes all the time consuming related to the tasks into account, including the platform's internal computation of qubit topology, the resource scheduling, the initialization of the qubits, the encoding of the parameterized quantum operations, the execution of the quantum circuits, etc.. They are sufficiently longer than the actual quantum circuit execution time. Consequently, we do not use the chip usage time directly for the speed comparing. Instead, we just treat them as part of the feedback of the tasks from Tianyan platform, proving the authenticity of the quantum circuit executions.

## Self-attention layer executions

Each repetition contains 24 successful tasks: 20 attention-head recovery segments, two mixing segments, and two softmax segments.

### Repetition 1

| Logical block | Executed segment | Task ID | Chip usage time |
|---|---|---|---|
| Head 0 | v | 2098943910413012994 | 6.725 s |
| | diag_minus | 2098943914565373953 | 7.148 s |
| | diag_plus_finish | 2098943919636287489 | 9.013 s |
| | selector | 2098943923927060481 | 10.553 s |
| | u | 2098943927198617602 | 7.507 s |
| Head 1 | v | 2098943931288064002 | 5.870 s |
| | diag_minus | 2098943935574605826 | 14.171 s |
| | diag_plus_finish | 2098943940557475842 | 13.521 s |
| | selector | 2098943944474955778 | 12.666 s |
| | u | 2098943947943608321 | 5.396 s |
| Head 2 | v | 2098943951919808514 | 0.308 s |
| | diag_minus | 2098943956067975170 | 9.576 s |
| | diag_plus_finish | 2098943961193451521 | 5.952 s |

| Logical block | Executed segment | Task ID | Chip usage time |
|---|---|---|---|
| | selector | 2098943965144449025 | 6.464 s |
| | u | 2098943968634109953 | 8.095 s |
| Head 3 | v | 2098943973067489281 | 8.156 s |
| | diag_minus | 2098943977643474946 | 10.089 s |
| | diag_plus_finish | 2098943982576013313 | 10.062 s |
| | selector | 2098943986749345794 | 5.350 s |
| | u | 2098943990046068737 | 11.712 s |
| Mixing | round_0 | 2098942612640829441 | 13.888 s |
| | round_1 | 2098942616281485314 | 4.945 s |
| Softmax | part_1 | 2098942620924542978 | 5.414 s |
| | part_2 | 2098942627501248514 | 10.125 s |
| Total | -- | -- | 202.706 s |

## Repetition 2

| Logical block | Executed segment | Task ID | Chip usage time |
|---|---|---|---|
| Head 0 | v | 2098943911247679490 | 6.725 s |
| | diag_minus | 2098943915827859457 | 7.148 s |
| | diag_plus_finish | 2098943920978464769 | 9.013 s |
| | selector | 2098943924623314946 | 10.553 s |
| | u | 2098943928347856898 | 7.507 s |
| Head 1 | v | 2098943932118536194 | 5.870 s |
| | diag_minus | 2098943936740622338 | 14.171 s |
| | diag_plus_finish | 2098943941673160705 | 13.521 s |
| | selector | 2098943945267679233 | 12.666 s |
| | u | 2098943948845383681 | 5.396 s |
| Head 2 | v | 2098943952955801601 | 0.308 s |
| | diag_minus | 2098943957213020161 | 9.576 s |
| | diag_plus_finish | 2098943962388828161 | 5.952 s |
| | selector | 2098943965924589570 | 6.464 s |
| | u | 2098943969720434690 | 8.095 s |
| Head 3 | v | 2098943974128648194 | 8.156 s |
| | diag_minus | 2098943978662690817 | 10.089 s |
| | diag_plus_finish | 2098943983859470337 | 10.062 s |
| | selector | 2098943987307188226 | 5.350 s |
| | u | 2098943991052701698 | 11.712 s |
| Mixing | round_0 | 2098942613416775681 | 13.888 s |
| | round_1 | 2098942617065820162 | 4.945 s |
| Softmax | part_1 | 2098942622564515841 | 5.414 s |
| | part_2 | 2098942629057335297 | 10.125 s |
| Total | -- | -- | 202.706 s |

## Repetition 3

| Logical block | Executed segment | Task ID | Chip usage time |
|---|---|---|---|
| Head 0 | v | 2098943911977488385 | 6.725 s |
| | diag_minus | 2098943917044207618 | 7.148 s |
| | diag_plus_finish | 2098943922190618626 | 9.013 s |
| | selector | 2098943925239877634 | 10.553 s |
| | u | 2098943929270603778 | 7.507 s |
| Head 1 | v | 2098943933016117249 | 5.870 s |
| | diag_minus | 2098943937906638850 | 14.171 s |
| | diag_plus_finish | 2098943942793039874 | 13.521 s |
| | selector | 2098943945921990658 | 12.666 s |
| | u | 2098943949831045122 | 5.396 s |
| Head 2 | v | 2098943953790468097 | 0.308 s |
| | diag_minus | 2098943958349676546 | 9.576 s |
| | diag_plus_finish | 2098943963663896577 | 5.952 s |
| | selector | 2098943966671175682 | 6.464 s |
| | u | 2098943970739650561 | 8.095 s |
| Head 3 | v | 2098943975089143809 | 8.156 s |
| | diag_minus | 2098943979895816193 | 10.089 s |
| | diag_plus_finish | 2098943985163898881 | 10.062 s |
| | selector | 2098943987965693954 | 5.350 s |
| | u | 2098943991916728322 | 11.712 s |
| Mixing | round_0 | 2098942614268219394 | 13.888 s |
| | round_1 | 2098942617917263873 | 4.945 s |
| Softmax | part_1 | 2098942624179322882 | 5.414 s |
| | part_2 | 2098942630672142337 | 10.125 s |
| Total | -- | -- | 202.706 s |

# FFN layer executions

Each repetition contains 10 successful tasks: three recovery segments for G_W1, two sigmoid-1 segments, three recovery segments for G_W2, and two sigmoid-2 segments.

## Repetition 1

| Logical block | Executed segment | Task ID | Chip usage time |
|---|---|---|---|
| G_W1 | v | 2098943994831732738 | 11.893 s |
| | diag | 2098943999588110338 | 5.385 s |
| | u | 2098944004373774337 | 8.401 s |
| Sigmoid 1 | part_1 | 2098942647415767041 | 13.392 s |
| | part_2 | 2098942653568847874 | 5.237 s |
| G_W2 | v | 2098944010803679233 | 8.730 s |
| | diag | 2098944015463550977 | 6.794 s |
| | u | 2098944020010139650 | 7.973 s |
| Sigmoid 2 | part_1 | 2098942672048914433 | 8.100 s |
| | part_2 | 2098942678080360450 | 7.577 s |
| Total | -- | -- | 83.482 s |

## Repetition 2

| Logical block | Executed segment | Task ID | Chip usage time |
|---|---|---|---|
| G_W1 | v | 2098943996157132801 | 11.893 s |
| | diag | 2098944000515051522 | 5.385 s |
| | u | 2098944006001164290 | 8.401 s |
| Sigmoid 1 | part_1 | 2098942648896356353 | 13.392 s |
| | part_2 | 2098942655187849218 | 5.237 s |
| G_W2 | v | 2098944012275879938 | 8.730 s |
| | diag | 2098944016281440258 | 6.794 s |
| | u | 2098944021624946689 | 7.973 s |
| Sigmoid 2 | part_1 | 2098942673495949314 | 8.100 s |
| | part_2 | 2098942679800025089 | 7.577 s |
| Total | -- | -- | 83.482 s |

## Repetition 3

| Logical block | Executed segment | Task ID | Chip usage time |
|---|---|---|---|
| G_W1 | v | 2098943997826465794 | 11.893 s |
| | diag | 2098944001391661057 | 5.385 s |
| | u | 2098944007745994753 | 8.401 s |
| Sigmoid 1 | part_1 | 2098942650473414658 | 13.392 s |
| | part_2 | 2098942656689410050 | 5.237 s |
| G_W2 | v | 2098944013660000257 | 8.730 s |
| | diag | 2098944017128689666 | 6.794 s |
| | u | 2098944023130701826 | 7.973 s |
| Sigmoid 2 | part_1 | 2098942675110756354 | 8.100 s |
| | part_2 | 2098942681607770114 | 7.577 s |
| Total | -- | -- | 83.482 s |